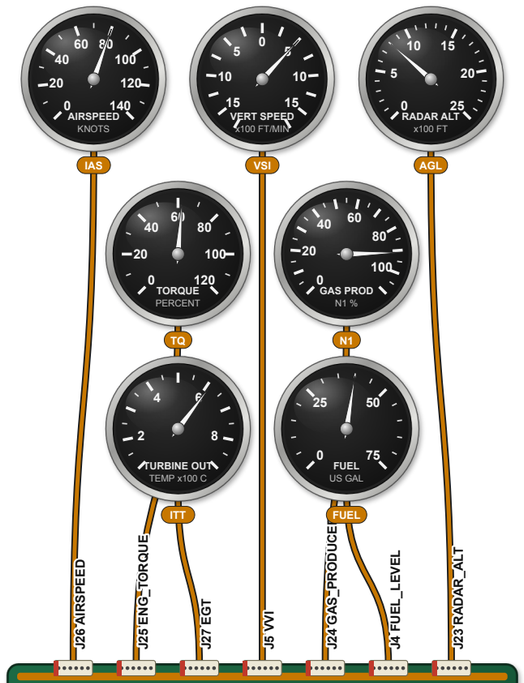
Jet Ranger cockpit: gauges, lamps and displays, and the controller board that drives each one

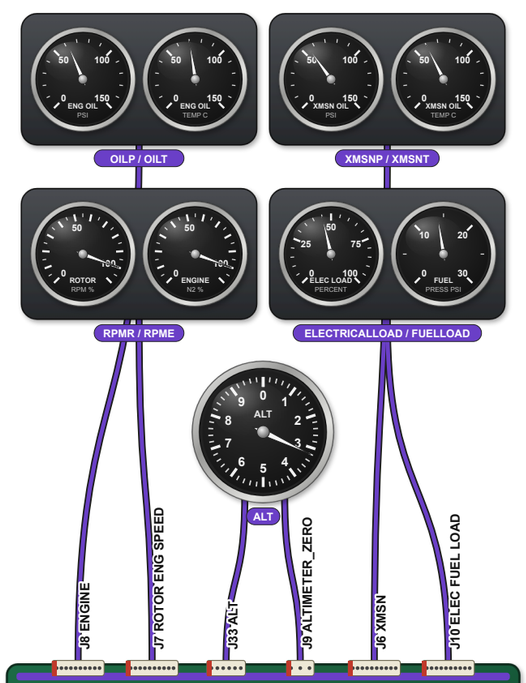
Instrument map, A3 landscape. Needles and lamps are drawn at illustrative readings. Cables end at the named connector on the board (references as printed on the PCBs).

STEPPER GAUGES (X27 steppers)



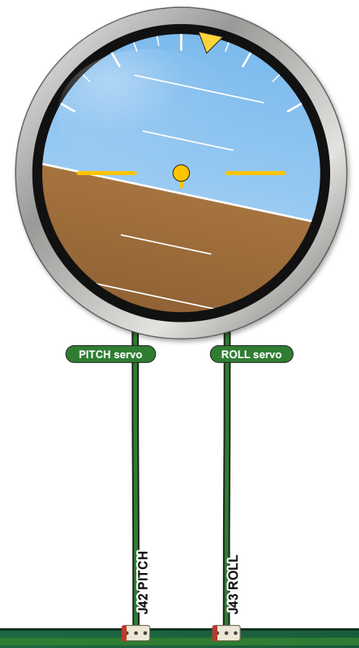
STEPPER CONTROLLER
172.16.1.105
JET_RANGER_STEPPER_CONTROLLER
Arduino Mega 2560 + W5500 | Jet Ranger 13 Stepper v1.6
ETHERNET 172.16.1.0/24

DUAL STEPPER GAUGES (dual instruments + altimeter)



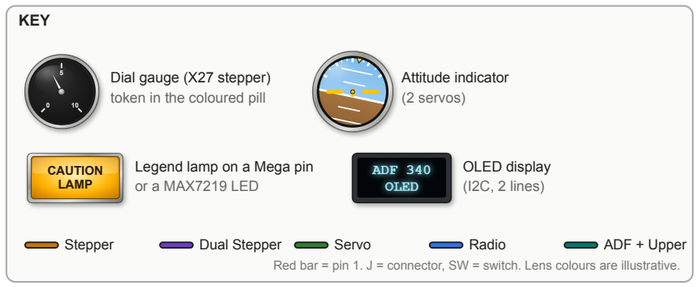
DUAL STEPPER CONTROLLER
172.16.1.106
JET_RANGER_OLED_DUAL_STEPPER_CONTROLLER
Arduino Mega 2560 + W5500 | OLED Dual Stepper v1.5
ETHERNET 172.16.1.0/24

ATTITUDE INDICATOR (pitch and bank servos)



SERVO CONTROLLER
172.16.1.102
JET_RANGER_SERVO_CONTROLLER
Arduino Mega 2560 + W5500 | Front Panel Servo v1.4
ETHERNET 172.16.1.0/24

KEY



HUEY WARNING PANEL (20 caution lamps on MAX7219 digits 4 - 7)

J20 - J23

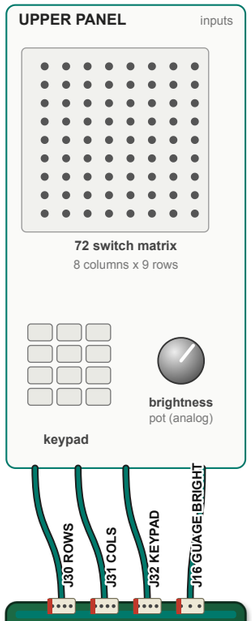
ENG OIL PRESS	ENG ICING	ENG ICE JET	ENG CHIP DET	L FUEL BOOST	R FUEL BOOST	ENG FUEL PUMP	20 MIN FUEL	FUEL FILTER	GOV EMER
CL_A0	CL_B0	CL_C0	CL_D0	CL_E0	CL_F0	CL_G0	CL_H0	CL_I0	CL_J0
AUX FUEL LOW	XMSN OIL PRESS	XMSN OIL HOT	HYD PRESS	INLET AIR	INST INVERTER	DC GENERATOR	EXT POWER	CHIP DETECTOR	IFF
CL_A1	CL_B1	CL_C1	CL_D1	CL_E1	CL_F1	CL_G1	CL_H1	CL_I1	CL_J1

J22 HUEY ANODES
J23 HUEY CATHODES
J20 HUEY SWITCHES

RADIO CONTROLLER
172.16.1.101
JET_RANGER_RADIO_CONTROLLER
Arduino Mega 2560 + W5500 | Radio Auto Pilot v1.4
ETHERNET 172.16.1.0/24

ADF PANEL

Other switches and knobs and the USB HID board are inputs: not drawn.



ADF + UPPER
172.16.1.107
JET_RANGER_ADF_CONTROLLER
Arduino Mega 2560 + W5500 | Upper ADF v1.0
ETHERNET 172.16.1.0/24

HOW THE VALUES ARRIVE

SIMULATOR
MSFS 2020 / 2024, Prepar3D, DCS World

↓

BRIDGE PROGRAM (on the PC)
FSUIPC, MSFS or DCS-BIOS bridge

↓

ETHERNET 172.16.1.0/24
UDP 13136: D,IAS:93.4,PITCH:4.5,...

↓

CONTROLLER BOARD
Mega 2560 + W5500, calibration tables

↓

CABLE TO THE CONNECTOR
stepper coils, servos, lamps, OLEDs

Each board takes real units and does its own calibration.

J55 BARO-OLED
J18 CLOCK-OLED
J17 ALTIMETER-OLED
J20 TEMPERATURE-OLED



DISPLAYS (4 OLEDs via a TCA9548A multiplexer) I2C D20 / D21

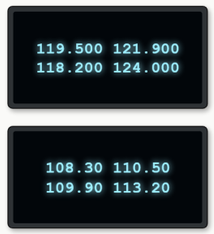
J4 LHS ANNUC
J5 RHS ANNUC

ANNUNCIATOR LAMPS (Mega pins A1 - A15)

ENGINE OUT	ROTOR LOW	XMSN OIL PRESS
EOUT A1	RLOW A2	TOPW A3
XMSN OIL TEMP	BATTERY HOT	BATTERY TEMP
TOTW A4	BHOT A5	BTMP A6
ENGINE CHIP	TAIL ROTOR CHIP	XMSN CHIP
EC A7	TRC A8	TC A9
BAGGAGE DOOR	FUEL PUMP	AFT FUEL FILTER
BD A10	FPMP A11	FFLTR A12
GEN FAIL	LOW FUEL	SC FAIL
GENF A13	LOWF A14	SCF A15

J8 COMM_NAV
J12 AUTOPILOT
J3 AUDIO
J5 ADF

COM AND NAV PANEL



COM and NAV frequency OLEDs (8 encoders and 4 swap switches are inputs, not drawn)

AUTO PILOT PANEL 8 LEDs



AUDIO PANEL 11 LEDs



ADF PANEL

